Technical Document

Digital Data Collection & Impact Monitoring Dashboard

Solar Powered Community Kitchen Initiative (SPCKI)

1. System Overview

The proposed system is a web / mobile-responsive digital data collection and monitoring platform for SPCKI. It will collect cooking, technical and household data, automatically calculate KPIs and present the results through an integrated dashboard.

Data Flow:

Data Collection → Central Database → KPI Calculation → Dashboard → Reports

2. Core Modules

1. **User management**
2. **Cooking Session Data**
3. **Daily Technical Data**
4. **Generator Maintenance**
5. **Monthly Household Feedback**
6. **KPI & Analytics**
7. **Impact Monitoring Dashboard**
8. **Reports & Data Export**
9. **Graphic Presentation**

User Management Software – Key Features

- User account creation, update & deactivation
- Role-based access control (RBAC)
- User permission & authorization management
- Secure login & password management
- Department/branch-wise user control
- User activity & audit trail tracking
- Login history monitoring
- User-wise and role-wise reports

3. Cooking Session Data

The system will record each household cooking session.

Field	Description
Date	Cooking date
Household ID	Participating household
Session	Session number
Start Time	Cooking start
End Time	Cooking end
Cooking Duration	Auto calculated
Food Type	Food category
Appliance	Appliance used
Cooking Completed	Yes/No
Waiting Issue	Yes/No
Technical Problem	Yes/No
Cleaning Completed	Yes/No
Remarks	Additional information

Automatic: Cooking duration, session status and relevant KPIs.

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4. Daily Technical Data

Field	Description
Date	Monitoring date
Daily Solar Production	kWh
Morning Battery SOC	%
Evening Battery SOC	%
System Operational	Yes/No
Technical Faults	Number
Total Downtime	Hours

Safety Incident	Yes/No
Weather Condition	Category
Remarks	Technical notes

5. Generator Maintenance

Field	Description
Date	Operation date
Fuel Feed	Litres
Running Time	Hours
Re-Fuel Feed	Litres
Maintenance Required	Yes/No
Maintenance Type	Type of maintenance
Remarks	Additional information

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6. Monthly Household Feedback

A short monthly survey will capture household experience and changes in backup-fuel use.

Indicator	Response
Backup Fuel Use	Yes/No
Backup Fuel Expenditure	Amount
Safety	1–5
Privacy	1–5
Waiting Experience	1–5
Overall Satisfaction	1–5
Kitchen Cleanliness	1–5
Ease of Cooking	1–5
Willingness to Continue	Yes/No
Remarks	Optional

Rating: 1 = Very Poor, 5 = Very Good.

7. Dashboard

A. Utilities

- Registered Households
- Active Households
- **Active Household Participation Rate**
- Total Cooking Sessions
- Completed Sessions
- **Session Completion Rate**

B. Cooking Performance

- Average Cooking Time
- Food Type Performance
- Appliance Utilization
- Session-wise Performance
- Incomplete Sessions
- Waiting Issues

C. Operational Performance

- Waiting Issue Rate
- Incomplete Session Rate
- Cleaning Compliance
- Technical Problems
- Common User Problems
- Safety Incidents

D. Technical Performance

- Daily/Monthly Solar Production
- Morning vs. Evening Battery SOC
- System Uptime
- Technical Faults
- Total Downtime
- Generator Running Hours
- Fuel Consumption

E. Household Feedback

- Backup Fuel Use

- Backup Fuel Expenditure
- Safety Score
- Privacy Score
- Waiting Experience
- Overall Satisfaction
- Kitchen Cleanliness
- Willingness to Continue

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8. Automatic KPI Calculation

KPI	Calculation
Participation Rate	Active HH ÷ Registered HH × 100
Session Completion Rate	Completed Sessions ÷ Total Sessions × 100
Average Cooking Time	Total Cooking Time ÷ Completed Sessions
Cleaning Compliance	Cleaned Sessions ÷ Total Sessions × 100
Waiting Issue Rate	Waiting Sessions ÷ Total Sessions × 100
Technical Fault Rate	Fault Sessions ÷ Total Sessions × 100
System Uptime	Operating Time ÷ Scheduled Time × 100
Average Downtime	Total Downtime ÷ Operational Days
Average Satisfaction	Total Rating ÷ Responses

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9. Dashboard Charts

The dashboard should provide simple visualizations:

1. Cooking Sessions – Daily/Weekly/Monthly
2. Solar Production Trend
3. Battery SOC Trend
4. Food Type Distribution
5. Appliance Utilization
6. Technical Fault & Uptime Trend
7. Waiting Issue Trend
8. Cleaning Compliance
9. Backup Fuel Use Trend
10. User Satisfaction Trend
11. Safety Incident Summary

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10. Data Validation & Security

The application should provide:

- Mandatory field validation
- Household ID validation
- Date/time validation
- Duplicate entry detection
- Automatic calculations
- Role-based access
- Secure login
- Audit trail
- Data backup
- Controlled data export

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11. Reports & Export

Daily

Cooking, solar, battery, faults, downtime and safety.

Monthly

Household participation, cooking performance, technical performance, generator operation, user feedback and backup-fuel use.

Export: Excel, CSV and PDF.

12. Recommended Technology Stack

Database
Oracle Database 19c / Oracle Database XE
Application / Frontend
Oracle APEX 24.2